

Pipe Flow Lab Report

Liem Fairbairn
02677394

Abstract

This report investigates the variance in results achieved through experimental and computational analysis of fluids, namely the coefficients of pressure and skin friction. The analysis is carried out on gaseous laminar flow at a Reynolds number of 1448, through a pipe of diameter 9.5mm using a centrifugal fan at the end of a pipe approximately 5m in length with atmospheric conditions in line with room temperature. It was found that computational results largely align with those of experimental and theoretical procedures, with discrepancy occurring largely due to modelling differences when executing the computational model.

Date 16th January 2026

Contents

List of Figures	2
List of Tables	2
List of Symbols	2
1. Introduction.....	3
2. Theory and Experimental Background	3
3. CFD Model and Setup	3
3.1 CFD Environment and CAD.....	3
3.2 Physics	4
4. Results and Discussions.....	5
4.1 – Coefficient of Pressure.....	5
4.2 – Coefficient of skin friction.....	6
5. Conclusion	7
6. References.....	8
7. Appendix.....	8
7.1 – Residual Plot	8
7.2 – Code	9
7.2.1 - MATLAB Code for finding the Coefficient of friction.....	9
7.2.2 – MATLAB Code plotting coefficient of pressure against distance along pipe.....	10
7.2.3 – MATLAB Code for Finding R^2 values in experimental and CFD data.....	10
7.3 – Tables	11
7.3.1 – Table 1: Data collected using a smoothed STAR-CCM plot of the velocity against radial distance from centreline:.....	11
7.3.2 – Table 2: Barometer height readings recorded during experimental procedure....	12

List of Figures

Figure 1 – Pressure Coefficient against Distance along Pipe.....	5
Figure 2 – Velocity against radial distance from centreline.....	6
Figure 3 – Experimental plot of C_f against Re.....	7
Figure 4 – Residual plot over 1000 iterations.....	8

List of Tables

Table 1 – Data collected using a smoothed STAR-CCM plot of the velocity against radial distance from centre line.....	11
Table 2 – Barometer height readings recorded during experimental procedure.....	12

List of Symbols

Re	Reynolds number
C_f	Skin friction coefficient
k	Orifice plate calibration factor
h	Barometer height reading (m)
h_0	Atmospheric barometer height (m)
x	Distance from tapping 1 (m)
D	Pipe diameter (m)
p	Pressure ($\text{kg m}^{-1} \text{s}^{-2}$)
C_p	Pressure coefficient
ρ_{air}	Density of air (kg m^{-3})
ρ_{alc}	Density of alcohol (kg m^{-3})
U_{inf}	Free stream flow velocity (m s^{-1})
U_{avg}	Mean/Bulk flow velocity (m s^{-1})
τ_{wall}	Shear wall stress ($\text{kg m}^{-1} \text{s}^{-2}$)
u	Horizontal flow velocity (m s^{-1})
y	Vertical displacement from centreline (m)
μ	Dynamic viscosity of air ($\text{kg m}^{-1} \text{s}^{-1}$)

1. Introduction

The investigation aims to compare the differences in results achieved through theoretical, experimental and computational methods of analysing pipe flow. Assessing the relationship between fluid properties and velocity is critical to the analysis of a body, as aspects such as skin friction and pressure coefficient dictate much of the energy dissipated through frictional forces and pressure drag respectively, thus having a big impact to the fuel efficiency of the aircraft, therefore contributing greatly to its environmental footprint, an ever-increasing point of importance for modern aircraft. Being able to model this computationally at a basic scale reduces cost of investigations by eliminating the need to carry out active wind tunnel testing and developing models, as well as the risk of human error in carrying out a manual experiment.

2. Theory and Experimental Background

The theoretical model with which we will compare our results is the following:

$$Re = \frac{16}{C_f} \quad (1)$$

This can be seen to be derived in by Daniel Joseph and Bobby Yang[1], achieved by assessing the development of a laminar boundary layer over a surface. Originally, the result was first put forward by Hagen and Poiseuille [2] in 1839, and it is their work which will act as the base comparative point of theory in this report.

We carried out an experimental procedure to recreate this result using pipe flow. By varying the voltage supply to a fan, we varied the velocity (calculated using an orifice plate) and recorded the change in height of barometer readings associated with seven pressure taps along the length of the tube. By analysing the relationship between the relative barometer heights and positions of the taps, we found the friction coefficient for each voltage supply using the following expression:

$$C_f = \frac{1}{4k} \times \frac{d(h/h_0)}{d(x/D)} \quad (2)$$

The changes in height were taken relative to the height at the atmospheric barometer height reading, and distances were taken relative to the first pressure tap.

3. CFD Model and Setup

3.1 CFD Environment and CAD

The CFD model made use of the STAR-CCM environment to simulate the pipe flow. A rectangular CAD model was loaded into the environment, to model a cross section of half of the pipe, which could be visualised as a 2D rectangle. This is due to the prism's rotational volumetric equivalence to the pipe, given the height of the CAD model was equal to the pipe radius. Thus, the height of the rectangle was 4.75mm, to match the pipe radius, and the length

of the model was approximately 5m to ensure fully developed flow, as the entrance length of the flow can be shown to be before the first tap (see figure 1) - implying a parabolic velocity profile via a constant pressure gradient. To the rectangular section modelling the pipe length, a quadrilateral mesh was applied of base size 0.001 metres, with a surface growth rate of 1.1, and the minimum surface size being equal to the target surface size, of 100%. One point of possible error in the CFD model, is the fact that the mesh used resembled a curvilinear mesh at its edges. This was ultimately ignored, however may be cause for a small source of error in using CFD outputs for calculation.

The rectangular prism's sides were allocated roles in the simulation, such as the two smallest faces (front and end) being allocated pressure outlet and velocity inlet respectively, and the top/side faces acting as a wall. The bottom face was to be a symmetry axis. The experimental model made use of a centrifugal fan to force air towards the fan, with airflow entering from the opposite side of the pipe. This meant that the pressure of fluid at the entrance of the pipe was atmospheric, that is, the relative pressure was set to 0 in the CFD model at the inlet. To achieve this effect computationally, the inlet at which flow entered the pipe was set to a pressure outlet, and inversely, the opposite end of the pipe was set to a velocity inlet. However, it was then necessary to then change the flow direction of the pipe to retain the expected fluid flow results.

3.2 Physics

The following assumptions regarding the fluid flow were made to simplify the CFD model:

1. Segregated flow – The effects of heat movement in the flow were not to be considered
2. Steady flow – The time effects on the fluid were to be ignored (fluid does not vary over time)
3. Constant Density – We are working in the laminar region, and thus the fluid can be modelled as incompressible.

Material properties of the gas were also accounted for, namely dynamic viscosity and density. The values $1.82 \times 10^{-5} \text{ (kg m}^{-1} \text{ s}^{-1}\text{)}$ [3], and $1.2 \text{ (kg m}^{-3}\text{)}$ [4] (note the value of 1.19 is used for CFD calculations) were used for dynamic viscosity and density respectively.

Within STAR-CCM, further options to specify a laminar gas flow were chosen to accurately represent the experiment. In all, this allowed for the modelling of a 2D axisymmetric incompressible, laminar and viscous fluid that is not expected to change over time. Further modifications to the model were made in the form of applying a no-slip condition to the walls, as well as initializing the initial conditions, setting the pressure of the inlet to 0 (relative to atmospheric), and the initial velocity to equal the velocity we attributed at our velocity inlet when assigning model roles to the CAD model. The residuals (see appendix 7.1) were shown to converge, thus validating the use of the CFD model detailed above.

4. Results and Discussions

4.1 – Coefficient of Pressure

When developing the CFD plot, it was necessary to use STAR-CCM to create an X-Y plot of pressure coefficient against distance, using the derived part ‘line probe’ to measure the pressure coefficient at different parts of the pipe. Every 5 points is shown in Figure 1 below for clarity. For our experimental results, plotting this required finding the pressure of the fluid by using barometer readings for 1.5V (see appendix 7.3.2.) using the following:

$$p = gh\rho_{alc} \times \sin(20) \quad (3)$$

$$C_p = \frac{p}{\frac{1}{2} \times \rho_{air} \times U_{inf}^2} \quad (4)$$

These coefficients were then plotted as co-ordinates, with the X co-ordinate being the distance of the tapping from tapping 1. It should be noted that the experimental plot shows the first reading occurring at 0m from the entry point of the pipe. This is not realistically the case. Shifting all co-ordinates by the distance tap 1 is from the opening will shift the plot towards the CFD result. This is evident in their approximately equivalent gradients (for calculation see appendix 7.2.2) of -4.6116 (experimental) and 4.4467 (CFD). This is a percentage change of 3.71% (3s.f.), which largely can be attributed to experimental error, likely random human error in data collection. Regarding experimental error, barometers had a resolution of 0.005 cm, and the thermometer 0.5°C.

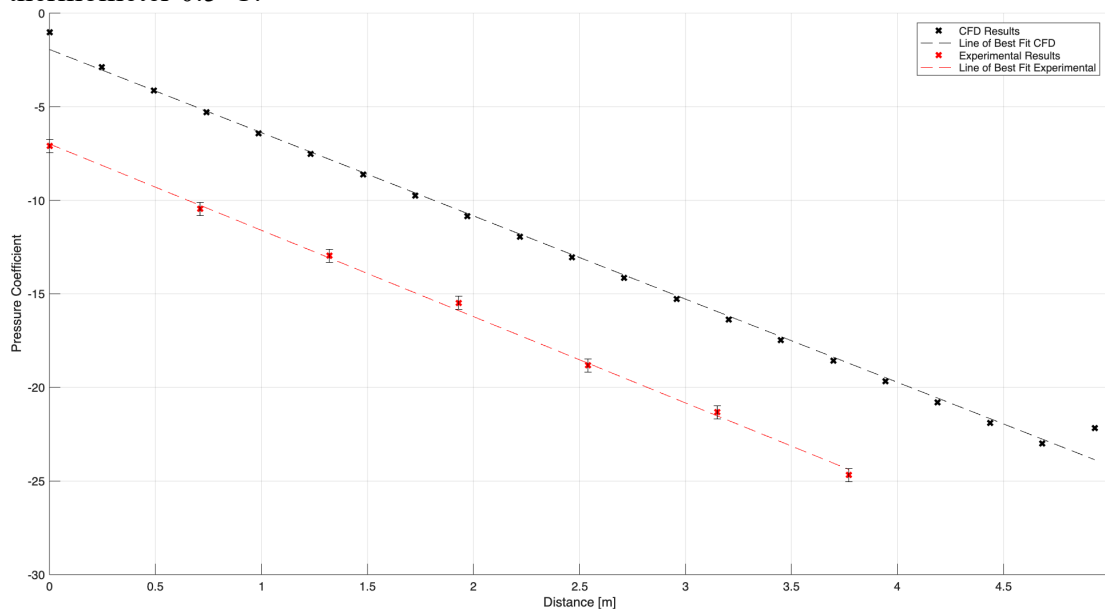


Figure 1 – Plot of Pressure Coefficient against distance along pipe

Error bars for experimental data derived using a percentage error in U_{inf} of 0.70327%, and a percentage error in p of 0.35%. These were found via errors in Δh , alongside the temperature reading in the orifice plate and standard taps along the pipe. No error bars present in CFD model. R^2 Values are 0.9984 (Experimental) and 0.9955 (CFD) – See appendix 7.2.3

4.2 – Coefficient of skin friction

The coefficient of skin friction is given by the following formula:

$$C_f = \frac{\tau_{wall}}{\frac{1}{2} \times \rho_{air} \times U_{avg}^2} \quad (5)$$

Where the shear stress at the wall is given as follows:

$$\tau_{wall} = \mu \times \frac{du}{dy} \quad (6)$$

The rate of change of velocity with respect of y was calculated by creating a probe line vertically across the model pipe cross section, measuring the horizontal velocity u at each probe. Within STAR-CCM, a data was then collected and developed in MATLAB (See appendix 7.2.1) to yield the following:

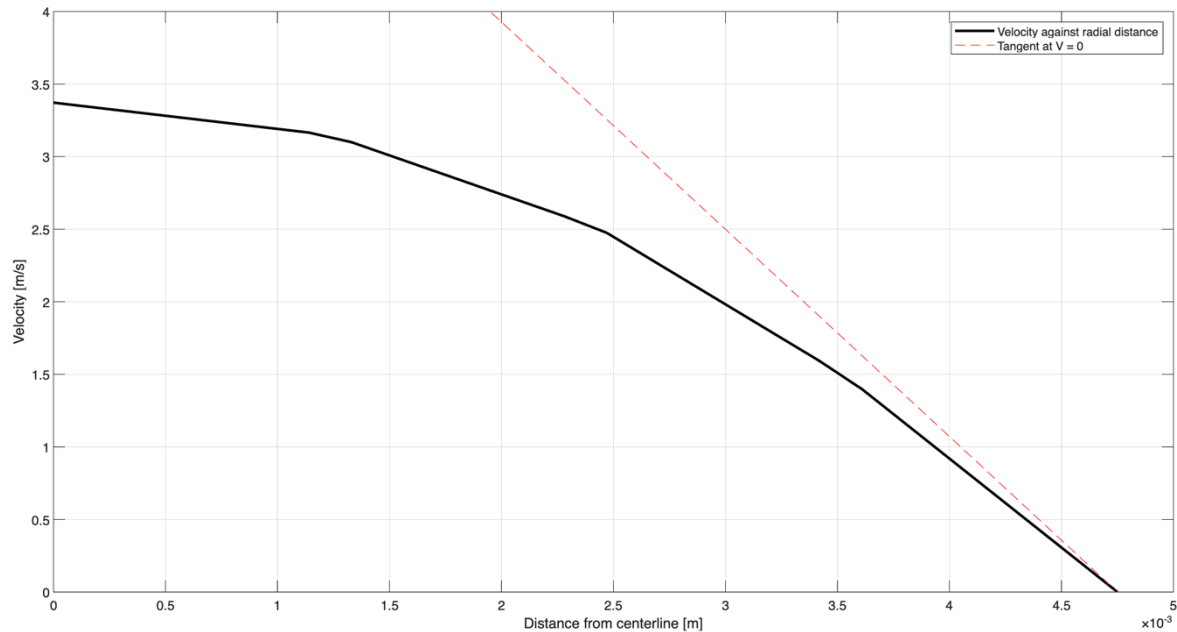


Figure 2 – Velocity against radial distance from centreline

The gradient was calculated across the final two points of data taken in the CFD model (see appendix 7.3.1), as it was concluded that a global quadratic fit was not appropriate, due to interest in only the near wall slope. This yielded a velocity gradient of -1.2258×10^3 , leading to the coefficient of skin friction being calculated as 0.0132. The value calculated by experiment was 0.011. This is a 20% error, which is likely attributed to perhaps a finite grid resolution in the CFD model. It is possible for misreadings of the height to have occurred, however we aimed to minimise this error with two group members reading each value to reduce the possibility of human error. It should be noted that data was smoothed within STAR-CCM, as our low mesh resolution only provided data for four points from the centreline to the wall.

This means that the penultimate data point used to calculate the gradient is approximately correct and appropriate for a relatively accurate calculation of the near wall gradient but not guaranteed to be wholly correct. The theoretical result [1] with which we will compare our findings is the following:

$$Re = \frac{16}{C_f} \quad (7)$$

Using this result, the Reynolds number of our pipe flow in CFD is calculated to be 1212 (4 s.f.), which is certainly still within the laminar region for our specified pipe, as we found experimentally (see figure 3). A percentage error of 20% sufficiently demonstrates consistency of experimental, computational, and theoretical approaches to assessment of fluid flow.

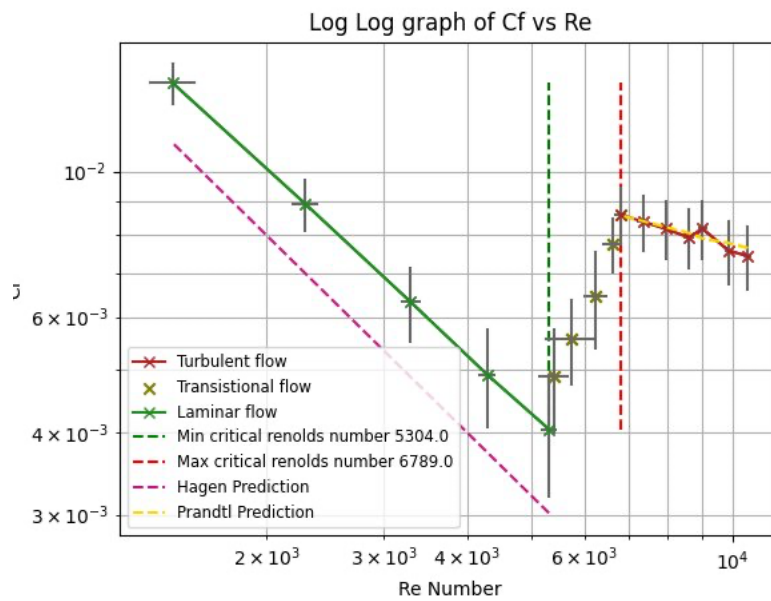


Figure 3 - Experimental plot of C_f against Re

Note: a plotting error was deduced to be the cause of inconsistency with theory, as experimental procedure was assessed to be correct using correctly collected and interpreted data.

5. Conclusion

Overall, despite error propagation in the computational and experimental models, the low discrepancy found in these procedures is indicative of their similarity. Comparatively, results differ to a significantly low degree to conclude both accuracy in the procedures taken, as well as the compatibility of computational modelling to represent experimental process. It can further be said that both computational models and experimental methods are accurate representations of known theoretical concepts in fluid mechanics and are able to verify the results of theory. This is ultimately supported by a difference of 3.71% in the gradients of the experimental and computational plots of pressure coefficient against pipe distance, as well as a 20% error in the coefficient of skin friction found in experiment, and the CFD model, where errors can be attributed to mesh geometry, idealisation of conditions within CFD, as well as the smoothing of data. It can be concluded, then, that there was a successful application of computational modelling to experimental and theoretical practice.

6. References

- [1] Joseph DD, Yang BH. Friction factor correlations for laminar, transition and turbulent flow in smooth pipes. *Physica D: Nonlinear Phenomena*. 2010; 239 (14): 1318–1328. 10.1016/j.physd.2009.09.026. [Last Accessed 16th January 2026]
- [2] Hagen. *Über die Bewegung des Wassers in engen cylinderischen Röhren*. *Annalen der Physik*. 1839. <https://onlinelibrary.wiley.com/doi/pdf/10.1002/andp.18391220304>. [Last Accessed 16th January 2026]
- [3] Katopodes ND. Chapter 1 - Basic Concepts. In: Katopodes ND. (ed.) *Free-Surface Flow*. Butterworth-Heinemann; 2019. pp. 2–98. [Last Accessed 16th January 2026]
- [4] Online. *The Engineering ToolBox (2003). Air Density, Specific Weight, and Thermal Expansion Coefficients at Varying Temperatures and Pressures*. https://www.engineeringtoolbox.com/air-density-specific-weight-d_600.html [Last Accessed 16th January 2026].

7. Appendix

7.1 – Residual Plot

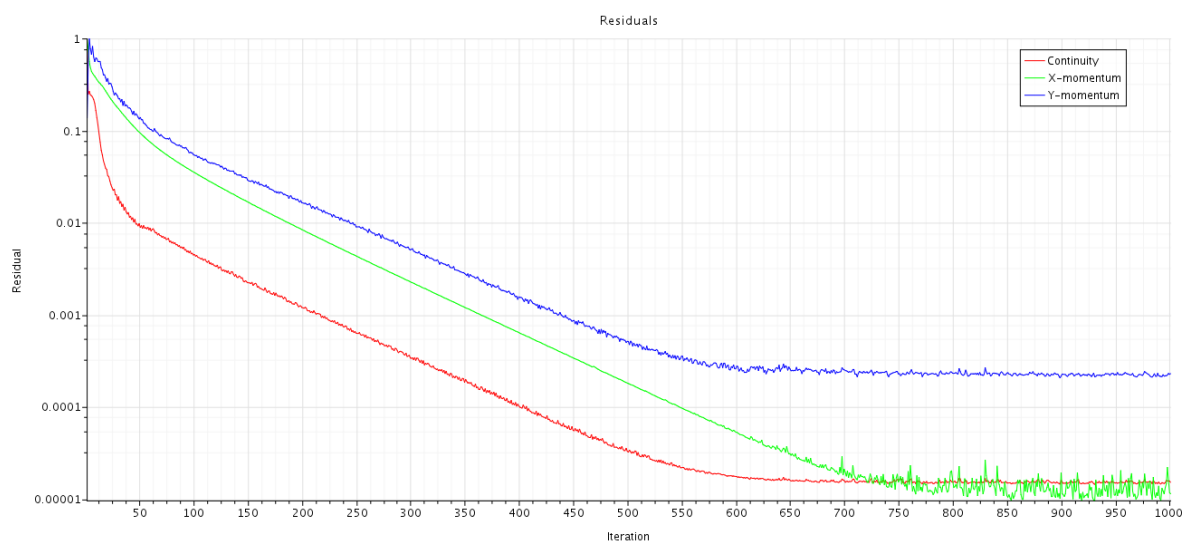


Figure 4 - Residual plot over 1000 iterations

7.2 – Code

7.2.1 - MATLAB Code for finding the Coefficient of friction

```
1 %Loading Velocity/Distance CFD data
2 data = readtable("VelData.xlsx");
3
4 velocities = data{1:end, 2}; %extracting data
5 distances = data{1:end, 1};
6
7 VEL_D = plot(distances, velocities, 'k', 'LineWidth', 2)
8 xlabel('Distance from centerline [m]')
9 ylabel('Velocity [m/s]')
10 grid on
11 hold on
12
13 %Mean Velocity
14 Vmax = velocities(1);
15
16 %Using the inlet velocity as the mean velocity
17 Vmean = Vmax/2
18
19 %Finding the value of the dU/dy at the wall
20
21 gradient = (velocities(end) - velocities(end-1)) / ...
22             (distances(end) - distances(end-1));
23
24 dU_by_dy = gradient
25 r=linspace(0,distances(end),length(distances))
26 V = gradient*(r-distances(end))
27
28 Tangent = plot(r,V, 'r--')
29 ylim([0 4])
30
31 legend([VEL_D Tangent], {'Velocity against radial distance', 'Tangent at V = 0'})
32
33 gradient = gradient*(-1);
34 %Calculating shear stress and then coefficient of friction
35 shear = 1.82*(10^(-5))*gradient;
36 C_f = shear/((0.5*1.19)*(Vmean^2))
```

7.2.2 – MATLAB Code plotting coefficient of pressure against distance along pipe

```

44 %1.5V pressure coefficient calculation
45
46 %equation is (P-Pinf)/(0.5*rho*Uinf^2)
47 %Constants
48 rho_air = 1.19;
49 %Pinf = 100960;
50 Uinf = 2.3223;
51 rho_barometer = 800;
52
53 %Modifying data collected in experiment
54 heights = [18.7 19.1 19.4 19.7 20.1 20.4 20.8]; %wrt atmosphere
55 heights = heights - mean([18.1 17.6]);
56 heights = heights.*10^(-2 ); %conversion to m
57
58 %calculating pressure of fluid
59 pressure = heights.*rho_barometer.*9.81.*sind(20);
60
61 %calculating coefficient of pressure
62 C_p = pressure/(0.5*rho_air*(Uinf^2));
63
64 %This represents positions of taps wrt Tap 1
65 xTAPS=[0, 0.71, 1.32, 1.93,2.54,3.15,3.77];
66
67
68 %need pressure for each tapping so we can find pressure C_p against X
69
70 %Loading data from CFD model
71 data2 = readtable("Pressurealongpipe.xlsx");
72 %The steps of 5 make data easier to express graphically
73 CoefficientP_CFD = data2{1:5:end, 2}; %extracting data
74 Distance2end = data2{1:5:end, 1};
75
76 %The suffix EXP represents 'experimental'
77 %The following defines the lines of best fit and the x axii values
78 linear_fitCFD = polyfit(Distance2end,CoefficientP_CFD,1);
79 linear_fitEXP = polyfit(xTAPS,-C_p,1);
80 x = Distance2end;
81
82 y = linear_fitCFD(1)*x + linear_fitCFD(2);
83 yEXP = linear_fitEXP(1)*xTAPS + linear_fitEXP(2);
84
85 %Plotting the graph
86 figure
87
88 CFD = scatter(Distance2end, CoefficientP_CFD, 'xk','LineWidth', 2);
89 hold on
90 LOBF = plot(x, y, 'k--');
91 hold on
92 EXP = scatter(xTAPS,-C_p,'xr','LineWidth',2)
93 hold on
94 LOBFEXP = plot(xTAPS, yEXP,'r--')
95 hold on
96
97 %producing an array to be used for error bars
98 for i = 1:7
99     percentage_array = (0.35/heights(i))*100+2*0.7037
100     Error_array = percentage_array.*(heights(i)/100)
101 end
102 errorbar(xTAPS,-C_p>Error_array)
103 hold off
104
105 %Labelling graph-Legend/Grid/Axii Limits
106 xlabel('Distance [m]')
107 ylabel('Pressure Coefficient')
108 legend([CFD LOBF EXP LOBFEXP], {'CFD Results', 'Line of Best Fit CFD','Experimental Results', 'Line of Best Fit Experimental'})
109 xlim([0 5])
110 ylim([-30 0])
111
112 grid on
113
114 %Finding Gradients
115
116 disp('The following are the gradeints for the experimental and CFD procedures respectively')
117 linear_fitEXP(1)
118 linear_fitCFD(1)

```

7.2.3 – MATLAB Code for Finding R^2 values in experimental and CFD data

```

120 %R^2 for experimental Data
121
122 p = polyfit(xTAPS,-C_p,1);
123 y_fit = polyval(p,xTAPS);
124
125 SS_residuals = sum((-C_p - y_fit).^2);
126 SS_total = sum((-C_p-mean(-C_p)).^2);
127 R_squared_EXP = 1 - SS_residuals/SS_total
128
129
130 %R^2 for CFD
131 p2 = polyfit(Distance2end,CoefficientP_CFD,1);
132 y_fit2 = polyval(p2,Distance2end);
133
134 SS_residuals2 = sum((CoefficientP_CFD - y_fit2).^2);
135 SS_total2 = sum((CoefficientP_CFD-mean(CoefficientP_CFD)).^2);
136 R_squared_CFD = 1 - SS_residuals2/SS_total2

```

7.3 – Tables

7.3.1 – Table 1: Data collected using a smoothed STAR-CCM plot of the velocity against radial distance from centreline:

Radial probe: Direction [0,0.00475,0] (m)	Radial probe: Velocity: Magnitude (m/s)
0	3.37
1.90E-04	3.34
3.80E-04	3.30
5.70E-04	3.27
7.60E-04	3.23
9.50E-04	3.20
0.00114	3.17
0.00133	3.10
0.00152	3.00
0.00171	2.90
0.0019	2.79
0.00209	2.69
0.00228	2.59
0.00247	2.47
0.00266	2.30
0.00285	2.12
0.00304	1.95
0.00323	1.77
0.00342	1.59
0.00361	1.40
0.0038	1.16
0.00399	0.933
0.00418	0.699
0.00437	0.466
0.00456	0.233
0.00475	0.00

7.3.2 – Table 2: Barometer height readings recorded during experimental procedure

Voltage	<u>manometers</u>										⊖
	a	1	2	3	4	5	6	7	8	9	
9	30.7 29.7 30.1 29.5	35.4	40.4 39.4	44.9	48.5	52.6	56.9	61.2	67.2	74.9	90
8.5	30.2 30.1	35.4	39.9	43.5	47.1	50.8	54.6	58.5	63.2	67.6	90
8	30.5 30.3	35.1	39.2	42.5	45.9	49.2	52.6	56.1	61.9	67.0	81
7.5	29.8 29.6	34.5	38.5	41.9	44.3	48.6	52.1	55.6	60.4	61.3	60
7	28.9 28.7	34.0	38.3	41.9	45.6	49.2	52.9	56.7	61.8	63.6	45
6.5	27.0 28.7	33.8	38.0	41.5	45.0	48.6	52.2	55.9	60.7	61.9	39.5
6	26.8 26.4	32	36.6	40.4	44.2	48.1	52.0	56.0	61.2	63.1	31
5.5	23.5 22.9	28.0 27.8	32.6 32.2	36.2 35.6	40.2 39.6	44.2 43.7	47.7 47.6	52.0 51.7	57.1 56.9	78.6 78.4	26
5	19.7 19.2	24.5 24.2	27.3 26.9	31.2 29.4	33.3 32.2	36.7 35.9	40.8 40.2	45.7 44.9	51 50.6	73.4 73.2	22
4.5	19.9 17.3	24.3 24.0	26.6 26.2	28.5 28.2	30.9 30.3	33.3 32.7	37.6 35.6	39.6 38.6	44.8 43.7	63.9 62.7	22
4	20.0 19.4	24.2 23.3	26.1 25.7	27.8 26.9	30.0 28.1	31.6 30.6	35.2 32.4	35.9 34.6	38.4 38.2	55.6 54.9	22
3.5	17.7 17.2	22.0	23.8	25.1	26.4	27.7	29.0	30.4	32.2	48.3	20
3	17.8 17.3	20.9	22.3	23.3	24.3	25.5	26.4	27.5	28.7	39.2	20
2.5	18 17.6	20.0	21.0	21.8	22.6	23.4	24.2	25.0	25.7	31.9	20
2	18.0 17.8	19.3 20.0	19.9 21.0	20.5 21.7	21.0 22.8	21.6 23.4	22.1 24.1	22.2 24.0	23 25.7	26 31.8	20
1.5	18.1 17.6	18.7	19.1	19.4	19.7	20.1	20.4	20.8	20.7	21.9	20
1											
0.5											